

REAL-TIME CREDIT CARD FRAUD RISK ENGINE

Edward Engelbert Wiguna



Real-Time Fraud Detection: Problem Background & Challenges

The Real-World Fraud Dilemma

- **Asymmetric Risk:** Direct loss of transaction amount + chargeback penalty fees
- **Friction vs. Security:** Overly aggressive blocking causes checkout friction and customer churn.
- **Failure of Default ML:** In a 0.15% fraud environment, a naive "never fraud" guess yields 99.85% accuracy but zero business utility.

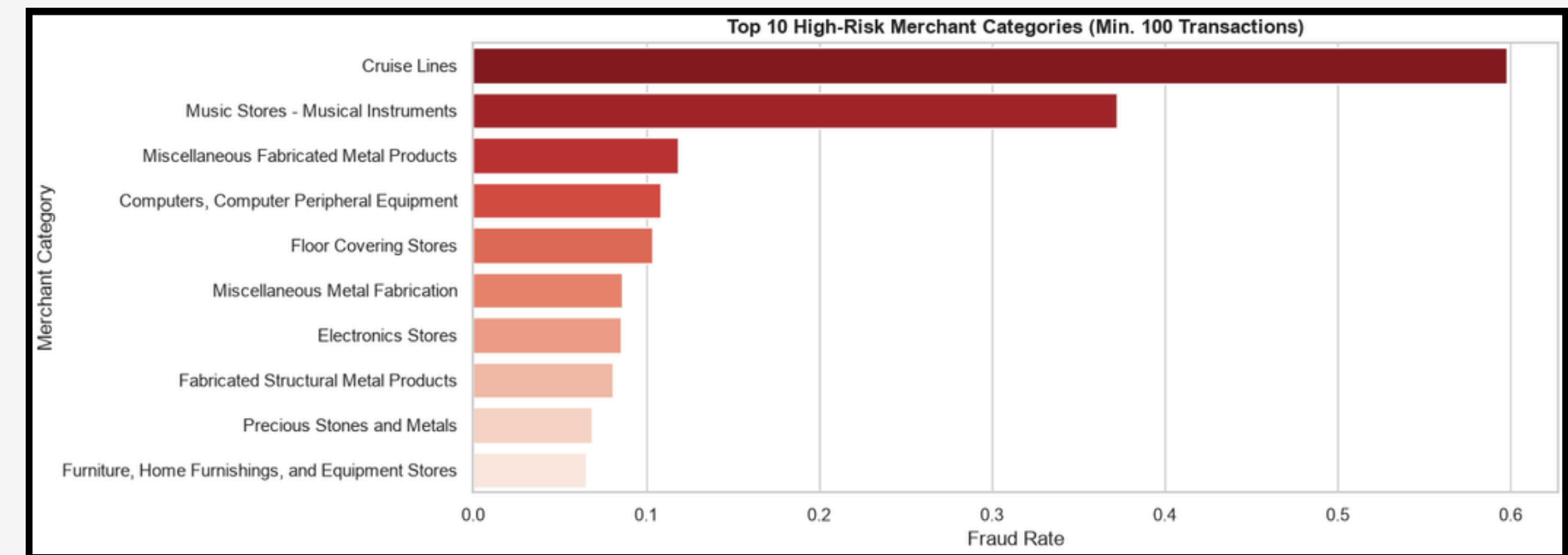
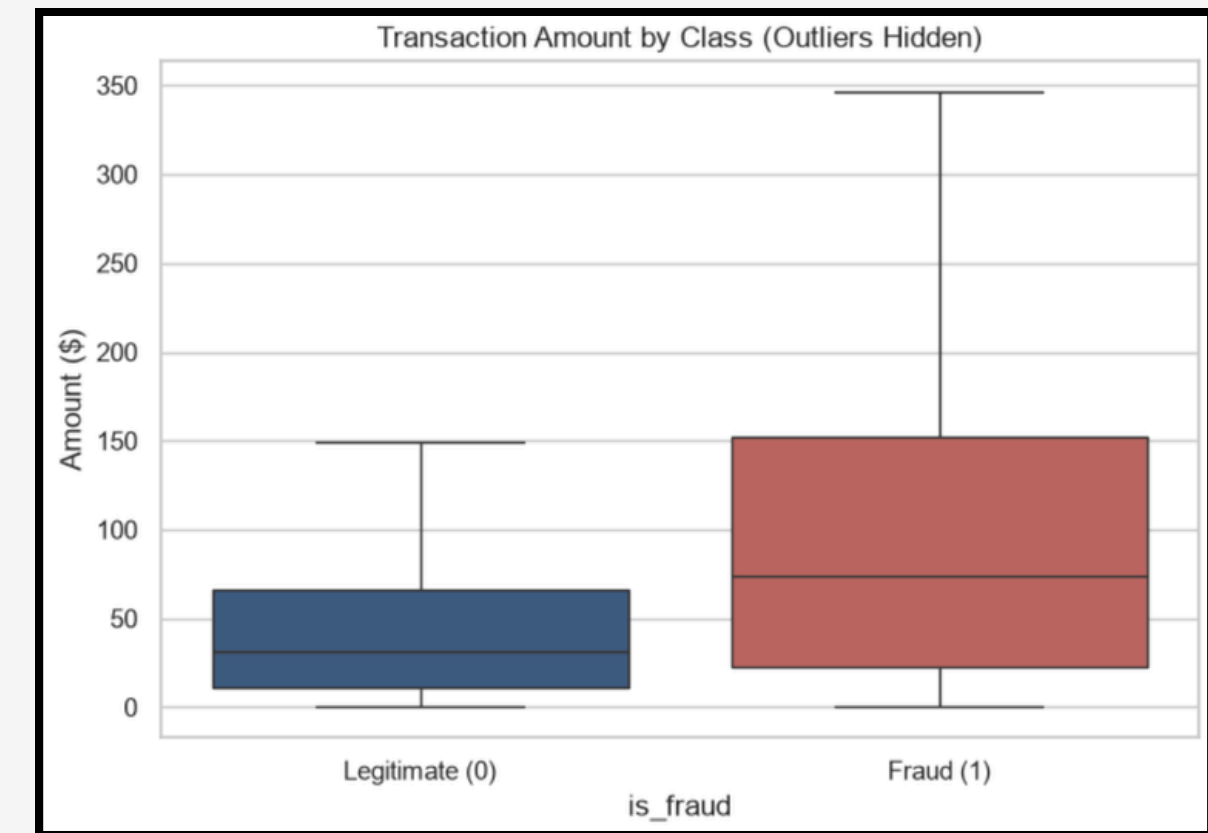
Dataset Scope & Characteristics

- **Severe Class Imbalance:** Base fraud rate of ~0.15% (~1 fraud per 667 transactions).
- **Relational Multi-Table Scope:** Transaction logs, user demographics, card metadata, and merchant categories.
- **Temporal Sensitivity:** Continuous, time-dependent stream requiring strict chronological handling to prevent leakage.

Exploratory Data Analysis & Diagnostic Insights

Key Findings

- **Higher Fraud Spend:** While legitimate spend is dominated by small everyday purchases, fraud leans heavily toward large ticket items.
- **Merchant Category Concentration:** Heavily concentrated in specific sectors like digital goods, electronics, and luxury retail.
- **Channel Vulnerability**



Dynamic Behavioral Feature Engineering

Velocity Burst ⚡

- `spend\_ratio\_1h\_to\_24h`
- Compares 1 hour activity against 24-hour baseline
- Identifies rapid-account draining behaviour

Merchant Deviation 🏪

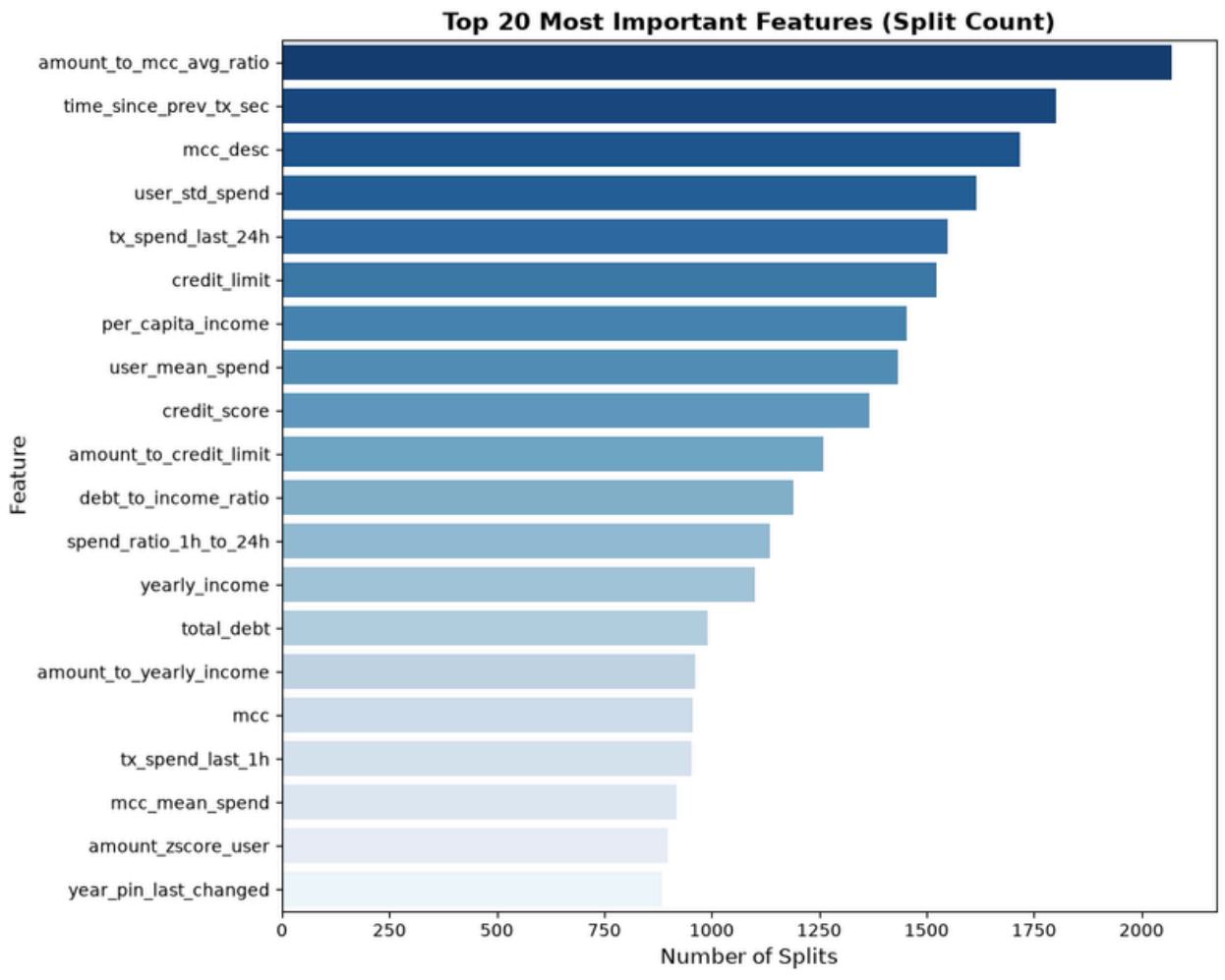
- `amount\_to\_mcc\_avg\_ratio`
- Compares spend against merchant category norm
- Flags unusual amount

Channel Exposure 🌐

- `online\_spend\_magnitude`
- Compares high value Card-Not-Present (CNP) e-commerce purchase
- Direct Risk Multiplier

Model Validation Results 📊

- Top Driver (#1): amount_to_mcc_avg_ratio is the single most important feature (>2,000 splits).
- Behavioral Dominance: The top 5 features are all spending patterns and timing, not static user profiles.
- Proves the Strategy: Dynamic velocity and merchant ratios easily outperform static demographics like age or location.



Model Development & Iterative Benchmarking

Baseline Model

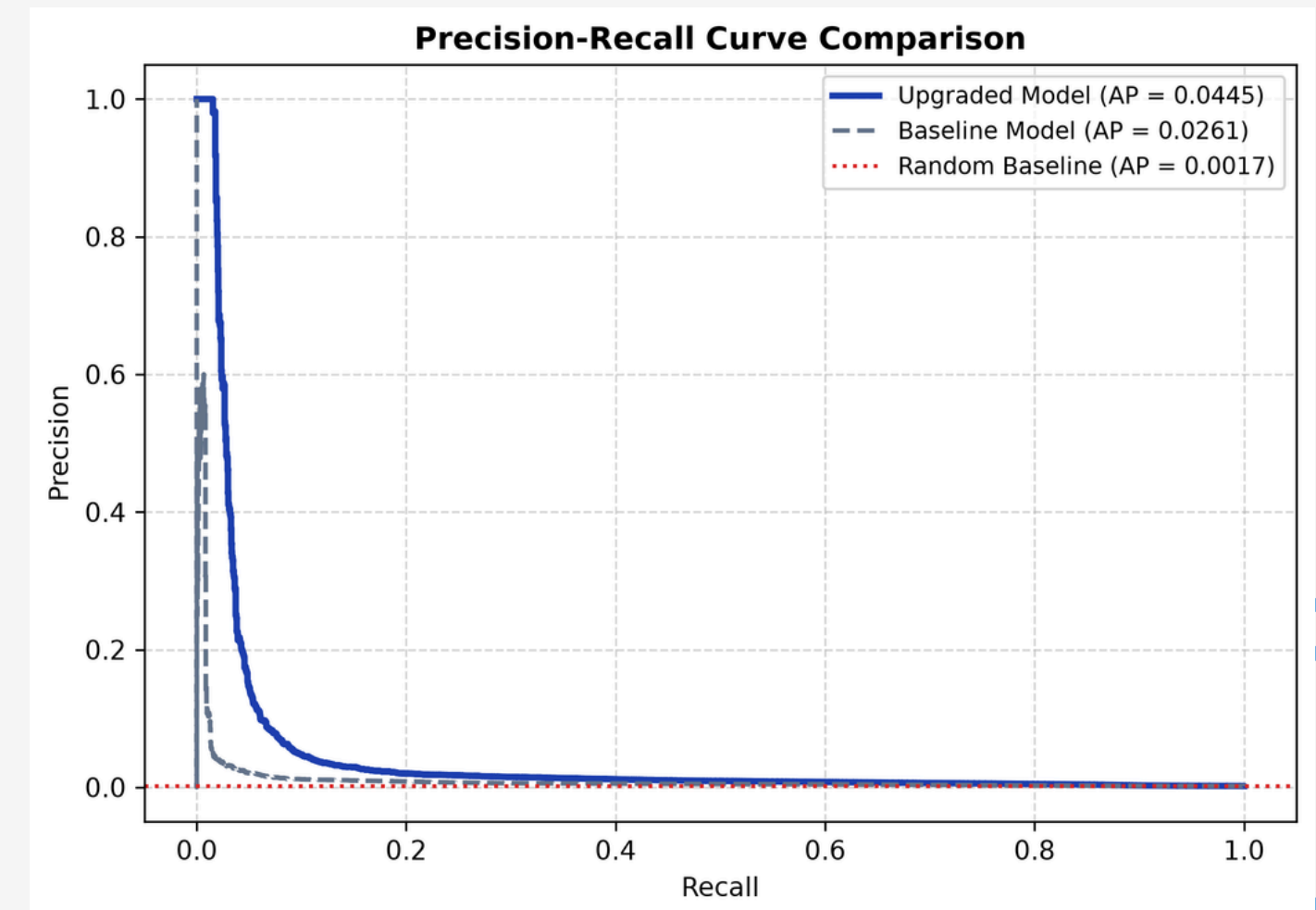
- Features: Raw fields + Demographics
- Hyperparameter: num_leaves: 63, Default
- Performance: **0.0261 PR-AUC** (~19x lift)

Upgraded Pipeline

- Features: Behavioral + Velocity ratios
- Hyperparams: num_leaves: 127, depth: 8
- Performance: **0.0445 PR-AUC** (~31x lift)

Key Takeaways

- KEY TAKEAWAY: +54% RELATIVE LIFT
- Behavioral features and tuned capacity boosted PR-AUC from 0.0261 to 0.0445.



Operational Decision Architecture (3-Tier Policy)

Tier 1: Auto-Approve (Score: < 0.05)

- Instant Authorization
- Covers ~97% of traffic
- Zero customer friction

Tier 2: Step-Up (Score: 0.05 - 0.35)

- Trigger 3DS / SMS OTP
- Verifies genuine users
- Resolves gray-zone spend

Tier 3: Decline (Score: > 0.35)

- Immediate Transaction Cut
 - Proactive card suspension
 - Near-zero false declines
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STRATEGIC BALANCE: Maximize fraud capture without hurting top-of-funnel conversion.

Financial Impact & Business ROI Simulation

\$48.5K+ 💰

Net Economic Benefit

- Net savings after OTP & Infrastructure costs



10.3 : 1

Verification ROI

- \$10.30 saved per \$1.00 spent on step-up checks.



99%+

Legitimate Approval Rate

- Preserves top-of-funnel checkout conversion



KEY TAKEAWAY: The 3-tier policy delivers high fraud prevention with near-zero friction for legitimate cardholders.

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=== FINANCIAL COST-BENEFIT SUMMARY ===
Total Fraud Exposure in Test Set : $291,295.19
Direct Fraud Saved (Hard Decline): $25,745.38
Assisted Fraud Saved (Step-Up)   : $28,019.92
Total Fraud Dollars Prevented     : $53,765.30 (18.46%)
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Operational Costs (OTP + Friction): $5,231.09
Net Economic Benefit Prevented    : $48,534.21
```

Next Steps & Production Rollout

1. Live Integration ⚡

- Plug model into checkout
- Silent shadow rollouts to verify live decisions

2. Monitor & Retrain 🔄

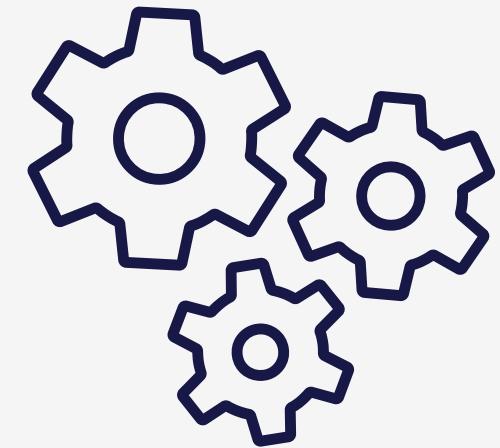
- Track accuracy weekly
- Update with new fraud examples over time

3. LLM / AI Assist 🤖

- Automatically turn technical data into easy-to-read fraud summaries.
- Send smart text alerts



KEY GOAL: Safely bring the model to life and support our team with smart AI tools.



Conclusion & Key Takeaways

High-Precision Model Performance 🚀

- +54% relative PR-AUC improvement (0.0445 score).
- ~31× lift over random baseline in critical risk windows.

Proven Bottom-Line Savings 💰

- \$48.5K+ net economic benefit generated in testing.
- High-value fraud intercepted before chargeback penalties hit.

Frictionless Customer Experience 🛡️

- Over 99% of legitimate checkout volume approved instantly.
- Smart text checks prevent false blocks

FINAL TAKEAWAY: An optimized risk engine that cuts fraud losses without hurting real customer sales.

THANK YOU

